UNIVERSITÉ DE GENÈVE

ANALYSE ET TRAITEMENT DE LINFORMATION
14X026

TP A : Linear Algebra

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1 Theory questions

Use the slides and your notes to answer these questions. You do not need to write code. Provide short explanations and examples where possible.

1. What is a vector? What is a vector space? Give an example of a vector in $\mathbb{R}^2$ or $\mathbb{R}^3$.

A vector is a mathematical object that has a length, a direction and a sense.

A vector space is a space of vectors with linear operations such as addition between vectors and multiplication by a scalar. It can be generated by the combination of a few vectors, according to the dimension of the space. For instance, we can generate all vectors of a vector space of dimension 2 by using 2 vectors.

2. What do we mean by the "dimension" D of a dataset?

The dimension of a dataset indicates the minimum number of vectors useful to generate the entire dataset by combinations between these vectors.

3. Create two small datasets: one scalar dataset of 5 numbers, and one vector dataset of 5 vectors in $\mathbb{R}^3$. Compute their mean and variance. Then, write the formal mathematical definitions of mean and variance.

The dataset are the following:

$$A = \{1,2,3,4,5\} \in \mathbb{R}$$

$$B = \left\{ \begin{pmatrix} 3 & 9 & 6 \\ 8 & 4 & 5 \\ 5 & 6 & 6 \\ 0 & 8 & 5 \\ 6 & 3 & 2 \end{pmatrix} \right\} \in \mathbb{R}^3$$

- The mean is given by the formula 1

$$\boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i \text{ for } \mathbf{x}_i, \boldsymbol{\mu} \in \mathbb{R}^d \tag{1}$$

So for the first dataset, we have the following computation:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} A_i = \frac{1+2+3+4+5}{5} = 3 \in \mathbb{R}$$

And for the second dataset, we have the following computation:

$$\boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n} B_i = \frac{1}{n}\begin{pmatrix}\sum_{i=1}^{n} B_{i1} & \sum_{i=1}^{n} B_{i2} & \sum_{i=1}^{n} B_{i3}\end{pmatrix} = \frac{1}{5}\begin{pmatrix}22 & 30 & 24\end{pmatrix} = \begin{pmatrix}4,4 & 6 & 4,8\end{pmatrix} \in \mathbb{R}^3$$

- The variance $\boldsymbol{\sigma}$ is given by the formula 2 by taking the square racin of the result

$$\sigma_j^2 = \frac{1}{n}\sum_{i=1}^{n} (x_{ij} - \mu_j)^2 \text{ for } \mathbf{x}_i, \boldsymbol{\sigma} \in \mathbb{R}^d \tag{2}$$

So we have for the first dataset

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n} (A_i - \mu)^2 = \frac{1}{5}\left((1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2\right) = \frac{10}{5} = 2$$

So $\sigma = \sqrt{2} \in \mathbb{R}$.

And for the second dataset

- j = 1

$$\sigma_1^2 = \frac{1}{n}\sum_{i=1}^{n} (x_{i1} - \mu_1)^2 = \frac{1}{5}\left((3-4.4)^2 + (8-4.4)^2 + (5-4.4)^2 + (0-4.4)^2 + (6-4.4)^2\right) = \frac{37.2}{5} = 7.44$$

- j = 2

$$\sigma_2^2 = \frac{1}{n}\sum_{i=1}^{n}(x_{i2}-\mu_2)^2 = \frac{1}{5}\left((9-6)^2+(4-6)^2+(6-6)^2+(8-6)^2+(3-6)^2\right) = \frac{1}{5}(9+4+0+4+9) = \frac{26}{5} = 5.2$$

- j = 3

$$\sigma_3^2 = \frac{1}{n}\sum_{i=1}^{n}(x_{i3}-\mu_3)^2 = \frac{1}{5}\left((6-4.8)^2+(5-4.8)^2+(6-4.8)^2+(5-4.8)^2+(2-4.8)^2\right) = \frac{10.8}{5} = 2.16$$

So

$$\boldsymbol{\sigma} = \begin{pmatrix}\sqrt{7.44} & \sqrt{5.2} & \sqrt{2.16}\end{pmatrix} \in \mathbb{R}^3$$

4. Define the scalar product. Compute it for two vectors from your vector dataset above. How does the scalar product allow us to compute the projection of one vector onto another?

The scalar product is defined for two vectors $\mathbf{x}$ and $\mathbf{y}$ like the formula 3

$$\mathbf{x}\cdot\mathbf{y} = \sum_{i=1}^{d} x_i y_i = |\mathbf{x}|\,|\mathbf{y}|\cos(\theta) \quad \forall \mathbf{x},\mathbf{y} \in S^d \tag{3}$$

with S the vector space of dimension d.

So for vector $\mathbf{x} = \begin{pmatrix}3 & 9 & 6\end{pmatrix}$ and $\mathbf{y} = \begin{pmatrix}8 & 4 & 5\end{pmatrix}$, the scalar product is given by the following calculs.

$$\mathbf{x}\cdot\mathbf{y} = \sum_{i=1}^{d} x_i y_i = 3\times 8 + 9\times 4 + 6\times 5 = 90$$

5. What is a basis of a vector space?

A basis of a vector space is a set of vectors that can generate the entire vector space by basics operations, such as addition between vectors and multiplication by scalars.

6. Explain the eigenvalue decomposition (EVD) and singular value decomposition (SVD). Why are they important in data science?

Write your answer here

7. Give one concrete example of the curse of dimensionality, and one example of the blessing of dimensionality.

Write your answer here

8. If you divide a D-dimensional space into b bins per dimension, and want k samples per bin, how many samples are needed in total?

Write your answer here

9. Why does data become more sparse as D grows? What happens if the number of samples stays fixed?

Write your answer here

10. Compare a hypersphere of radius r with its enclosing hypercube $[-r, r]^D$. As D increases, what happens to their volume ratio? What does this imply for uniform sampling?

Write your answer here

11. In high dimensions, what is the typical angle between two random vectors? How does this angle change as D increases?

Write your answer here

12. What is meant by a "concentration phenomenon"? Give one example where it is useful, and one where it is problematic.

Write your answer here

13. State one concentration inequality (e.g., Markov, Chebyshev, Hoeffding). Explain briefly what it tells us.

Write your answer here

14. What is a PAC (Probably Approximately Correct) analysis in machine learning?

Write your answer here

2 Matrix

1. Find a quadratic polynomial, say $f(x) = ax^2 + bx + c$, such that

$$f(1) = -5, \quad f(2) = 1, \quad f(3) = 11.$$

Write your solution here

2. Let a, b be some fixed parameters. Solve the system of linear equations

$$\begin{cases} x + ay = 2 \\ bx + 2y = 3 \end{cases}$$

Write your solution here

3. Find the inverse of the following matrix, if it exists:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

Write your solution here

4. Solve the following matrix equation for (X):

$$AX = B, \quad \text{where} \quad A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \end{pmatrix}.$$

Write your solution here

3 The importance of the mathematical concept behind a code

1. Download, open and run several times the PYTHON file "some_script.py".

   Explain the function `def project_on_first(u, v)` in geometrical terms and then explain the results of this code with clear mathematical concepts (you have to talk about projections and scalar product).

   Write your solution here

2. Consider $u = (u_i)_{1 \leq i \leq 7}$ and $v = (v_i)_{1 \leq i \leq 7}$, two vectors of length 7. How would you clearly write the following part of the script using some mathematical notation?

```python
import numpy as np
u = np.random.randn(7)
v = np.random.randn(7)
r = 0
for ui, vi in zip(u, v):
    r += ui * vi
```

   Name the mathematics behind and rewrite the last 3 lines of code in one.

   Write your solution here

3. Complete the code to construct a vector v orthogonal to the vector u and with the same norm as u. Comment each line of your code.

   Write your solution here

4. Given two vectors (u) and (v) in $\mathbb{R}^n$, write Python code to compute the cosine of the angle between them. Explain the mathematical concept behind this computation, and show how you would write it in one equation.

   Write your solution here

4 Computing Eigenvalues, Eigenvectors, and Determinants

1. Find the determinant, eigenvectors, and eigenvalues of the matrix

$$\begin{bmatrix} 5 & 6 & 3 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \end{bmatrix}$$

   Compare your results with those of the computer.

   Write your solution here

2. The covariance matrix for n samples $x_1, \ldots, x_n$, each represented by a $d \times 1$ column vector, is given by

$$C = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \mu)(x_i - \mu)^T,$$

   where C is a $d \times d$ matrix and $\mu = \sum_{i=1}^{n} x_i / n$ is the sample mean.

   Prove that C is always positive semidefinite. (Note: A symmetric matrix C of size $d \times d$ is positive semidefinite if $v^T C v \geq 0$ for every $d \times 1$ vector v.)

   Write your solution here

3. In this portion of the exercise, we will calculate the eigenvalues of the covariance matrices of six data sets listed as follows:

filename	n	d	description
tp1_artificialdata[1-3]	1024	100	Artificial data generated from various auto-regression
tp1_artificialdata4	1024	100	Random Gaussian data
tp1_freyfaces	1965	560	Facial images of a man named Brendan
tp1_digit2	5958	784	Hand-written images of "2"

Download and unzip `data for python.zip`. Each file contains a $n \times d$ data matrix with rows representing n different samples in $\mathbb{R}^d$.

(a) Compute the covariance matrix of each dataset.

(b) Compute the eigenvalues for each covariance matrix.

(c) Compute the determinant of each covariance matrix.

(d) Compute the product of the eigenvalues for each covariance matrix.

(e) Display the eigenspectrum for each covariance matrix in a 2D plot.

Describe the relationship between the product of the eigenvalues and the determinant of each covariance matrix. In addition, describe the observations you made regarding the plot of the spectrum of eigenvalues of each covariance matrix. How can you explain the disparities between datasets?

Write your solution here

4. Let $A \in \mathbb{R}^{n\times n}$ be a square matrix. Show that if (A) has an eigenvalue (λ= 0), then the matrix (A) is singular. Furthermore, prove that the determinant of (A) must be zero if any eigenvalue is zero.

Write your solution here

5 Computing Projection Onto a Line

There are given a line $\alpha : 3x + 4y = -6$ and a point A with the coordinates $(-1, 3)$.

1. Find a distance from the point A to the line α using linear algebra (not coordinate method).
2. Explain your code using a sketch and some mathematics (there should be a scalar product somewhere).

`_Write your solution here_`